A Zero-Parameter Framework for Cosmological Dynamics and Galaxy Cluster Mass Discrepancies via Structural Self-Energy Expansion (SSEE-V3.6)

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Abstract

We introduce the Structural Self-Energy Expansion (SSEE-V3.6), a purely algebraic framework that proposes a connection between cosmological expansion and galaxy-cluster dynamics using only two dimensionless geometric constants — π and the golden ratio φ — with a minimal effective parameter count and no WIMP, QCD-axion, or SUSY dark-matter particles. The background sector $(w_0, w_a, \Omega_{\rm DE}, \Omega_{m,\rm dyn})$ is fully algebraic (zero fitted dimensionless parameters); phenomenological extensions in the dark-matter sector (Paper 6) introduce a small set of quantities currently treated as ansätze and tracked as open problems (OP-8, OP-9, OP-11, OP-14 in `OPEN_PROBLEMS.md`). From these two axioms, a closed set of seven structural registers is derived algebraically, culminating in the dark-energy equation of state

$$w_0 = -\frac{3\varphi+\pi}{2(\varphi+\pi)} \approx -0.840, \qquad w_a = -\frac{2\varphi+\pi}{2(\varphi+\pi)} \approx -0.670.$$

No fitting procedure is involved: w_0 and w_a are uniquely fixed by the geometry of φ and π. The asymptotic structural viscosity factor $\mathrm{KAL}_0 = \beta + \pi \approx 5.5214$ natively reproduces the observed cosmic dark-to-baryonic matter density ratio. These foundational parameters are tested against data in subsequent papers: a full Bayesian MCMC validation against DESI DR2 [Abdul-Karim et al., 2025], Planck 2018 [Planck Collaboration VI, 2020], and galaxy-cluster mass data ($\Delta\mathrm{BIC} = -5.55$ for the dynamic sector relative to ΛCDM) is presented in the companion Paper 2 [Almeida Vallejo, 2026], while CMB perturbations and phenomenological extensions are explored in Papers 3–7. The framework is falsifiable: a B-mode polarisation tensor-to-scalar ratio $r = \varphi^{-10} = 0.00813$ is predicted for LiteBIRD [LiteBIRD Collaboration, 2023] ($\sim$2032).

1 Introduction and Theoretical Foundations

The observed cosmic acceleration [Perlmutter et al., 1999, Riess et al., 1998] requires an explanation beyond the cosmological constant [Weinberg, 1989], motivating a broad

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landscape of dynamical dark energy proposals [Ratra & Peebles, 1988, Copeland et al., 2006, Weinberg et al., 2013, Clifton et al., 2012]. The SSEE framework operates under the principle of algebraic closure and macroscopic structural self-energy. Rather than adding particle degrees of freedom, the system's identity is defined by a closed set of geometric invariants derived exclusively from π and φ. These fundamental scales dictate the macroscopic behaviour of spacetime:

- $\boldsymbol{\Omega}$ ($\pi + \varphi \approx 4.7596$): Stability Metric.
- $\boldsymbol{\beta}$ ($(\pi + \varphi)/2 \approx 2.3798$): Base Coupling Scalar.
- $\mathbf{KAL_0}$ ($\beta + \pi \approx 5.5214$): Asymptotic Structural Viscosity.
- $\boldsymbol{P}$ ($\Omega + \varphi \approx 6.3776$): Dynamical Evolution Scalar.
- $\boldsymbol{K_v}$ ($\varphi + \pi + \Omega \approx 9.5192$): Structural Constraint Invariant.
- $\boldsymbol{T_r}$ ($3(\varphi + \beta) \approx 11.9935$): 3D Saturation Horizon.
- $\boldsymbol{M_v}$ ($\varphi + \pi + K_v \approx 14.2788$): Maximal Dimensional Invariant.

1.1 Predictive Register

A common objection to algebraic models is selection bias: that the constant combinations are chosen post-hoc to reproduce known values. Table 1 documents the chronological sequence of the SSEE derivations relative to the observational data they are compared against. The Genesis 5.12 compendium (Zenodo DOI: `10.5281/zenodo.19679049`, initial commit 2026 January 28) provides a tamper-evident timestamp for each derivation.

Table 1: Predictive Register: SSEE derivations in chronological order. "Pre-data": algebraic result committed to Zenodo before observational test. "Prediction": not yet observationally constrained. "Retrodiction": derived independently, then compared to existing measurement. No optimisation over the algebraic formula space was performed.

Quantity (algebraic formula)	Value	Test / Falsifier	Status
Papers 1–2 (Genesis 5.12, Zenodo 2026-Jan-28)			
$w_0 = -T_r/M_v$	-0.8399	DESI DR2 (2025)	Pre-data
$w_a = -(\Omega+\varphi)/K_v$	-0.6699	DESI DR2 (2025)	Pre-data
MIRA $= (3\varphi+\pi)/4$	1.9989	$\Omega_{m,\mathrm{CMB}}/\Omega_{m,\mathrm{dyn}}$	Pre-data
$\Omega_m = (\pi-\varphi)/(\pi+\varphi)$	0.3201	Planck 2018	Retrodiction
$n_s = 1-\varphi^{-7}$	0.96556	Planck 2018	Retrodiction
$H_0 = 3(\varphi+\pi)^2$	$67.96\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$	Planck 2018	Retrodiction
Paper 5 — IS causal perturbation theory			
$\gamma_{\mathrm{IS}} = 0.554$	$\approx \gamma_{\Lambda\mathrm{CDM}} = 0.55$	RSD surveys	Retrodiction
$f\sigma_8(z{=}0.5)$	mean 0.74σ vs 6 RSD surveys (Paper 5 corrected)	BOSS/eBOSS/6dFGS	Retrodiction
Paper 6 — φ-Dark Matter, two-sector			
$m_\phi = \Sigma m_\nu^{\mathrm{active}} \times H_0^{\mathrm{alg}}$	5.60 eV	Matter $P(k)$ via k_{fs} (no SM portal)	Prediction (ansatz)
k_{fs} (Dodelson-Widrow from m_ϕ)	$0.493\,h\,\mathrm{Mpc}^{-1}$	DESI Y3 / Euclid $P(k)$, 2026–2028	Prediction
Paper 7 — Canonical EFT, Bellini-Sawicki α-functions			
$\alpha_T = \alpha_M = \alpha_B = 0$ (minimal coupling)	0 exact	GW170817 $\lvert\alpha_T\rvert < 10^{-15}$	Retrodiction
$\alpha_K = 3\,\Omega_{\mathrm{DE}}(1+w_\phi)$	0.073	Euclid weak lensing, 2026–2028	Prediction
$\beta_c = -\mathrm{AURA} = -(3\varphi+\pi)/2$	-3.9978	EFT plateau test (8 ICs, $\Delta = 0.2\%$)	Retrodiction
Future			
$r = \varphi^{-10}$	0.00813	LiteBIRD $\sim$2032	Prediction

"Retrodiction" denotes results derived independently and then compared against previously published values; no optimisation over the algebraic formula space was performed to match a target value. The distinction between pre-data predictions and retrodictions is explicitly noted to maintain transparency. All entries with "Prediction" status are falsifiable by named experiments within the stated time horizon; a single null detection would rule out the SSEE framework.

1.2 Axiomatic Principles

SSEE is explicitly constructed as a top-down theoretical ansatz. Its internal symmetries are not phenomenological fits derived bottom-up, but fundamental axiomatic identities that the macroscopic universe is proposed to satisfy. Three physical postulates govern the interactions of these geometric invariants:

1. **The Temporal Triad (Cosmological Epochs):** Spacetime evolves through three distinct thermodynamic states — the Origin/Law phase (initial singularity), the Ignition/Processing phase (structure formation), and the Transcendental Saturation phase (dark energy asymptotic domination). The Hubble history $H(a)$ is bounded by:
$$\lim_{a\to 0} H^2(a) \propto \rho_{\rm bar}, \qquad \lim_{a\to\infty} H^2(a) = \frac{8\pi G}{3}\,\rho_{\rm crit,0}\left(\frac{T_r}{M_v}\right). \tag{1}$$
2. **Non-Self-Addition (Evolution by Difference):** Structural evolution proceeds through non-linear geometric scaling rather than linear superposition. All macroscopic observables $\mathcal{O}$ must be strictly functional dependencies of the invariant set and H_0; no additional free parameters are permitted:
$$\mathcal{O} = f(\Omega,\, \beta,\, \mathrm{KAL}_0,\, P,\, K_v,\, T_r,\, M_v;\, H_0). \tag{2}$$
3. **Asymmetric Equivalence:** Energy states can possess identical geometric values but divergent dynamic functions. The macroscopic geometric response must inversely mirror the physical source:
$$\mu_{\rm SSEE}(x)\cdot \mathrm{KAL}(x) \equiv 1. \tag{3}$$

1.3 Scope of the Zero-Parameter Claim

The invariant set $\{\Omega, \beta, \mathrm{KAL}_0, P, K_v, T_r, M_v\}$ and its generators φ, π are all *dimensionless*. Two foundational postulates make precise what the framework derives and what it takes as input; both are stated here as pre-registered axioms rather than left implicit.

Postulate D (Dimensional Anchor). Because the invariant set is dimensionless, SSEE predicts only *dimensionless* quantities: equation-of-state parameters (w_0, w_a), density fractions $(\Omega_{\rm DE}, \Omega_m)$, the EFT functions $(\alpha_K, \alpha_B, \alpha_M, \alpha_T)$, the spectral observables (n_s, r), and the structural ratios (MIRA, etc.). These are unit-independent and directly falsifiable. Any *dimensionful* quantity — H_0, $\rho_{\rm crit}$, particle masses — requires exactly *one* dimensional input, the anchor, for which we take H_0. This is the identical status H_0 holds in ΛCDM, where it is one free parameter. The "zero free parameters" claim therefore means precisely: *zero dimensionless fitted parameters.* SSEE carries one dimensional input (H_0) and zero dimensionless ones; ΛCDM carries one dimensional input (H_0) and five dimensionless ones $(\Omega_m, \Omega_b, n_s, A_s, \tau)$. The algebraic combination $3(\varphi+\pi)^2 \approx 67.96$ numerically coincides with H_0 expressed in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$; consistent with the result-type protocol (§C) this is recorded as a *Type-P numerical coincidence*, not a derivation of the dimensional scale. Deriving the absolute scale of H_0 from a fundamental scale is the cosmological-constant problem — the ratio $H_0/M_{\rm Pl} \sim 10^{-61}$ is not bridged by any closed algebraic form — and lies beyond SSEE as it does beyond every current framework.

Postulate S (Saturation Correspondence). Define the *saturation fraction*

$$s \;\equiv\; \frac{T_r}{M_v} \;\in\; (0,1). \tag{4}$$

The dark sector is a one-parameter family in s, with the density fraction and the equation of state both fixed by it:

$$\Omega_{\rm DE} = s, \qquad w_0 = -s. \tag{5}$$

The two endpoints are physically forced and require no assumption:

- $s \to 0$: $(\Omega_{\rm DE}, w_0) \to (0,0)$ — the Einstein–de Sitter (matter-only) limit;
- $s \to 1$: $(\Omega_{\rm DE}, w_0) \to (1,-1)$ — the exact de Sitter limit, already imposed by Postulate 1.

An immediate consequence is the much-used identity

$$1 + w_0 \;=\; 1 - s \;=\; \Omega_{m,\rm dyn}, \tag{6}$$

which is therefore *not* an independent assumption but the content of Postulate S. We state explicitly what remains a postulate rather than a theorem: the endpoints fix $(0,0)$ and $(1,-1)$, but a nonlinear assignment $w_0 = -s^n$ $(n \neq 1)$ would meet the same endpoints. Linearity $(n = 1)$ is the minimal, lowest-order correspondence and is adopted as such; deriving $n = 1$ from a dynamical attractor of the Israel–Stewart sector is identified as open theoretical work.

Auxiliary structural postulates beyond D and S. A careful audit of the derivation chain (this paper plus the companion Papers 3–10) reveals that beyond Postulates D and S, the framework rests on two additional auxiliary postulates that do not follow automatically from D+S and are introduced as algebraic-register declarations:

- **Postulate M (MIRA).** The ratio between the CMB-sector and the dynamical-sector matter densities is the structural constant MIRA $= (3\varphi+\pi)/4 = \text{AURA}/2 \simeq 1.9989$. The *value* is algebraic; the *dynamical mechanism* that realises this ratio in a Friedmann background remains an open problem (OP-8 in `OPEN_PROBLEMS.md`; seven candidate mechanisms have been ruled out). The alternative geometric form $\Omega_{m,\rm CMB}^{\rm geom} = (\pi - \varphi)/(\pi + \varphi) \simeq 0.3201$, recorded in the Predictive Register, is an *independent* algebraic prediction (not a separate postulate, since it uses no new structural ingredients): it agrees with the MIRA-mapped prediction MIRA$\cdot\Omega_{m,\rm dyn} \simeq 0.3199$ to 0.054% and both lie well within 1σ of Planck 2018. The near-coincidence between two independent algebraic routes is treated as evidence in favour of the register structure, not as a discrepancy to be repaired.
- **Postulate I (Inflationary register).** The inflationary sector is an α-attractor with $\alpha = \varphi^4/3$ and the horizon-crossing e-fold number has the algebraic form $N_* = 2\varphi^n$ for some integer n. What this postulate fixes is the *ansatz form* (α-attractor

class, the specific algebraic value of α, the structural form $2\varphi^n$), not the integer n itself. The value $n = 7$ is a corollary, not an assumption: among integer exponents, $N_* = 2\varphi^n$ falls inside the canonical inflation window $[50, 60]$ uniquely for $n = 7$ ($2\varphi^6 = 35.9$, $2\varphi^7 = 58.07$, $2\varphi^8 = 94.0$). Equivalently, the exponent is selected by an observational constraint external to SSEE (the standard inflation duration), not by an internal numerological choice. From these inputs follow the standard α-attractor relations $n_s = 1 - 2/N_* = 1 - \varphi^{-7}$ (matches Planck PR4 at 0.16σ) and $r = 12\alpha/N_*^2 = \varphi^{-10}$ (LiteBIRD prediction). Neither the α-attractor class, the value $\alpha = \varphi^4/3$, nor the form $N_* = 2\varphi^n$ is derived from D+S; together they constitute Postulate I.

The honest count is therefore **two foundational postulates (D, S) plus two auxiliary register-level postulates (M, I)** — four total, all dimensionless. The geometric form $(\pi-\varphi)/(\pi+\varphi)$ is an additional algebraic prediction within the same register system, not a fifth postulate. The integer $n = 7$ in Postulate I is also not part of the assumption budget — it is a corollary of the ansatz form combined with the canonical inflation window. Together with the open problems OP-8/9/11/14 for the dark-matter sector, these define the full assumption budget of SSEE-V3.6.

Honest accounting of effective parameters. The zero-parameter status established above applies strictly to the *background* sector ($w_0, w_a, \Omega_{\mathrm{DE}}, \Omega_{m,\mathrm{dyn}}$) and to the dimensionless EFT functions of Paper 7. Phenomenological extensions introduce a small number of effective parameters that, while assigned algebraic values in the current version, do not yet possess a derivation from first principles and are tracked as open problems:

- The MIRA factor $(3\varphi+\pi)/4$ has an algebraic value but no derived dynamical mechanism (OP-8). Seven candidate mechanisms have been ruled out (cs^2 clustering, $\mu > 1$ Poisson, disformal P8, conformal $\beta_c = -\mathcal{A}$, saturated backward, derivative coupling, forward φ-MDE).
- The φ-DM mass $m_\phi = 5.60\,\mathrm{eV}$ in Paper 6 is a numerological ansatz (OP-9), depending on Σm_ν, itself classified as Type P (phenomenological) in Paper 4 due to the integer offset in $\mathcal{R} = 4\,\mathrm{KAL} - 22$ (OP-14).
- The non-minimal coupling ξ of φ-DM is currently a free parameter (OP-11).

The framework therefore carries $\sim$3 effective dimensionless ansätze beyond the strictly algebraic background, against six fitted parameters in ΛCDM ($\Omega_m, \Omega_b, n_s, A_s, \tau, H_0$). Resolution of OP-8/9/11/14 would restore the literal zero-parameter status; see `OPEN_PROBLEMS.md` for the dependency chain.

1.4 The Two-Ω_m Criterion

A persistent question — both internal and from referees — is the use of two distinct matter density values across the suite: $\Omega_{m,\mathrm{dyn}} = 0.160$ and $\Omega_{m,\mathrm{CMB}} = 0.320$, linked by the MIRA factor $(3\varphi + \pi)/4 \simeq 1.9989$. These are not free parameters: they are the same physical content evaluated on either side of a single free-streaming scale, $k_{\mathrm{fs}} = 0.493\,h\,\mathrm{Mpc}^{-1}$, derived in Paper 6 from the ϕ-DM mass ansatz $m_\phi = 5.60$ eV via the Dodelson–Widrow relation. The criterion for which value enters a given observable is therefore physical and pre-registered:

Observable / context	Ω_m	**Reason**
Paper 2 — late-time geometry, BAO, clusters:		
DESI DR2 BAO chain, low-z $E(z)$	0.160	local geometry, $k \gg k_{\rm fs}$.
r_d inside the BAO chain	0.160	enters $E(z)$-consistent late-time mapping.
Galaxy-cluster dynamical mass	0.160	non-linear scales, ϕ-DM unbound.
Paper 3 — CMB Boltzmann:		
CMB peak positions ℓ_n	0.320	$D_A(z_*)$ integrates ϕ-DM, non-rel. at z_*.
r_d from full CAMB Boltzmann	0.320	ϕ-DM non-rel. at recomb.
Paper 4 — algebraic identities:		
$\Omega_{m,\rm CMB} = (\pi-\varphi)/(\pi+\varphi)$	0.320	CMB-sector closed-form identity.
$\Omega_{m,\rm dyn} = (\pi-\varphi)/[2(\varphi+\pi)]$	0.160	dyn-sector identity (= CMB/MIRA).
Tensor-to-scalar $r = \varphi^{-10}$	–	primordial, no Ω_m dependence.
Paper 5 — Israel–Stewart growth:		
SSEE Friedmann background $E(z)$	0.160	dynamical sector, $1 + w_0 = \Omega_{m,\rm dyn}$.
Poisson source in linear growth ODE	0.320	full matter on $k < k_{\rm fs}$ scales.
$\Omega_{m,\rm eff} = \Omega_{m,\rm dyn} + \Omega_{\rm DE} r$ (clustering)	0.160+	DE clustering deviation, $r \ll 1$.
Paper 6 — two-sector dark matter:		
$\Omega_{\rm CDM}$ (always active)	0.160	dynamical CDM, all k.
$\Omega_{\phi\rm DM} = (\text{MIRA} - 1)\Omega_{m,\rm dyn}$	0.160	active only $k < k_{\rm fs}$.
$\Omega_{m,\rm growth} = \Omega_{\rm CDM} + \Omega_{\phi\rm DM}$	0.320	sum on clustering scales.
Paper 7 — EFT canonical:		
EFT slope $\lambda = \sqrt{3\,\Omega_{m,\rm dyn}} \approx 0.693$	0.160	background scalar dynamics.
$\alpha_K(0) = 3\Omega_{\rm DE}(1 + w_0)$	0.160	$1 + w_0 = \Omega_{m,\rm dyn}$ identity.
Growth ratio $G_{2s} = D_1^{\rm SSEE}(\Omega_m{=}0.320)/D_1^{\Lambda\rm CDM}$	0.320	two-sector growth comparison.
Paper 8 — strong gravity, lensing:		
Small-scale photon deflection, CDM only	0.160	only CDM clusters at $k \gg k_{\rm fs}$.
Cluster/LSS lensing total, $\Omega_{m,\rm eff} = \Omega_{\rm CDM} + \Omega_{\phi\rm DM}$	0.320	both sectors on $k < k_{\rm fs}$.
Paper 9 — Hubble tension:		
Late-time local CPL background $E(z)$	0.160	local dynamical sector.
AURA cancellation identity $f_{\rm scr} = (\pi-\varphi)/(\varphi+\pi)^2$	–	algebraic, no Ω_m argument.
Paper 10 — UV completion:		
$\alpha_K^{\rm full}/(3\,\text{MIRA})$, $r_V \propto 1/M^2$	–	uses MIRA ratio only; background Ω_m unchanged.

The criterion is therefore: any observable whose relevant comoving scales lie *below* $k_{\rm fs}^{-1}$ (i.e. probing $k > k_{\rm fs}$) at low redshift uses $\Omega_{m,\rm dyn} = 0.160$; any observable that integrates over scales $k < k_{\rm fs}$ *or* originates in the early universe where ϕ-DM is already non-relativistic and contributes to the matter density uses $\Omega_{m,\mathrm{CMB}} = 0.320$. The MIRA factor is the algebraic statement that the ratio between the two is fixed and not tunable.

The dual appearance of r_d in the table reflects the consistent application of this rule, not a duplication. The BAO chain in Paper 2 tests the late-time $E(z)$ geometry with $\Omega_{m,\mathrm{dyn}} = 0.160$ and uses an r_d value consistent with that background; the CAMB Boltzmann integration in Paper 3 computes the sound horizon from the recombination physics where ϕ-DM is already non-relativistic and therefore enters with the full $\Omega_{m,\mathrm{CMB}}\, h^2$. Matching the Planck-measured $r_d \simeq 147$ Mpc in *both* contexts is a non-trivial consistency test of the MIRA mapping.

Two independent algebraic predictions for $\Omega_{m,\mathrm{CMB}}$. The CMB-sector matter density admits two independent algebraic expressions within the SSEE register system:

$$\Omega^{\mathrm{geom}}_{m,\mathrm{CMB}} = \frac{\pi - \varphi}{\pi + \varphi} = 0.320100 \quad \text{(Paper 4, YGG\_PROTON form)}, \tag{7}$$

$$\Omega^{\mathrm{MIRA}}_{m,\mathrm{CMB}} = \mathrm{MIRA} \cdot \Omega_{m,\mathrm{dyn}} = \frac{3\varphi + \pi}{4} \cdot \frac{\pi - \varphi}{2(\varphi + \pi)} = 0.319928 \quad \text{(Paper 3, MIRA mapping)}. \tag{8}$$

These are two distinct algebraic operations on $\{\varphi, \pi\}$ and there is no a priori reason for them to coincide — they are not the same expression. Numerically they differ by 5.4×10^{-4} (0.054%), and both lie well within 1σ of the Planck 2018 measurement $\Omega_m = 0.3153 \pm 0.0073$. The near-coincidence between two structurally independent algebraic routes is recorded in the Predictive Register as cross-confirmation of the register system, not as a discrepancy to be repaired. Postulate M (MIRA) is the load-bearing assumption for the CMB sector; the YGG_PROTON form is an independent algebraic prediction that happens to land at the same observed value, which is evidence for the structural choice of registers.

Because k_{fs} is itself derived from m_ϕ (Paper 6), which is currently a phenomenological ansatz (OP-9, depending on OP-14), the *value* of k_{fs} inherits the same status. The *criterion*, however — the physical principle that a free-streaming scale separates the two regimes — is structural, not adjustable: it would hold for any species whose free-streaming length lies between the BAO and the CMB acoustic scales.

1.5 The ssee Effective Action

Instead of an arbitrary cosmological constant Λ, the vacuum energy density is fixed by the algebraic ratio T_r/M_v, leading to the background-level effective action:

$$S^{\mathrm{bg}}_{\mathrm{SSEE}} = \int d^4x\, \sqrt{-g} \left[\frac{R}{16\pi G} - \rho_{\mathrm{crit},0} \left(\frac{T_r}{M_v} \right) + \mathcal{L}_{\mathrm{matter}} \right]. \tag{9}$$

This action encodes the background Friedmann dynamics only: the dark-energy density is constant at the background level and set by the algebraic ratio T_r/M_v. It does not propagate scalar perturbations; the perturbation-level dynamics require a scalar-field extension, developed in the EFT embedding of Appendix A (Eq. 10). Within the Horndeski class of scalar-tensor theories [Horndeski, 1974, Gubitosi et al., 2013, Bellini & Sawicki, 2015], a k-essence scalar [Armendariz-Picon et al., 2001] with Lagrangian $K(X) = X/\mathrm{KAL}_0$ reproduces w_0 algebraically while remaining ghost-free and gradient-stable ($c_s^2 = 1$; Papers 7 and 10).

Falsifiability. The framework is strictly falsifiable. It would be refuted by: (1) direct detection of a collisionless dark-matter particle accounting for the missing mass; (2) definitive cosmological constraints proving $w_a = 0$; or (3) high-significance confirmation of a normal neutrino hierarchy with $\sum m_\nu < 0.06\,\mathrm{eV}$.

2 Effective Field Theory Embedding

Scope and limitations of this section. The background dynamics presented here use the *Eckart first-order* approximation [Eckart, 1940], which is sufficient to derive w_0, w_a, and the viscosity coupling KAL_0 on superhorizon scales. The Eckart formulation is known to violate causality in the relativistic perturbation sector [Müller, 1967, Hiscock & Lindblom, 1983, 1985]; for perturbation-level predictions (B-mode forecasts, sound-speed constraints) the Israel–Stewart (IS) formulation is required and is developed in Paper 5. The Eckart and IS frameworks agree at leading order in the slow-roll expansion, so all background results below are unchanged by the IS extension.

Sound speed caveat (EFT embedding). For a pure power-law k-essence Lagrangian $P(X) = A\,X^n$ the adiabatic sound speed satisfies $c_s^2 = 1/(2n-1)$. With $n \approx -0.095$ (derived below) this gives $c_s^2 \approx -0.84 < 0$, indicating gradient instability in the *power-law IR approximation*. This instability is an artefact of the $P(X) \propto X^n$ ansatz and is resolved in Papers 7 and 10 by replacing the power-law with the canonical k-essence Lagrangian [Armendariz-Picon et al., 2001] $K(X) = X/\mathrm{KAL}_0$ (gradient-stable, $c_s^2 = 1$) and its UV completion $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$ ($c_s^2 \in (1/3, 1)$ globally). Both replacements reproduce w_0 algebraically via $w_\phi = K/(2XK_X - K)$ without requiring $n < 0$. The power-law sector presented in this appendix should be regarded as a *motivational framework* for the algebraic structure of w_0, w_a at the background level; the perturbation-stable formulation is developed in Papers 7 and 10.

2.1 The SSEE Action

The SSEE dark-energy sector is described by a k-essence scalar field ϕ minimally coupled to gravity and extended by a geometric bulk-viscosity term in the Eckart approximation [Eckart, 1940, Weinberg, 1971]:

$$S = \int d^4x\,\sqrt{-g}\left[\frac{R}{16\pi G} + P(X) - \zeta\left(\nabla_\mu u^\mu\right)^2 + \mathcal{L}_m\right], \tag{10}$$

where $X \equiv -\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi$ is the canonical kinetic term. The fluid four-velocity of the dark-energy component is identified with the normalised gradient of the field,

$$u_\mu = -\frac{\partial_\mu\phi}{\sqrt{2X}}, \tag{11}$$

so that $u_\mu u^\mu = -1$ is satisfied automatically, and the viscous and kinetic sectors remain self-consistently defined by the single degree of freedom ϕ.

2.2 Algebraic Determination of $P(X)$

For a power-law k-essence Lagrangian $P(X) = \mathcal{A}X^n$ the equation of state is *exactly* constant:

$$w_\phi = \frac{P}{2XP_X - P} = \frac{\mathcal{A}X^n}{(2n-1)\mathcal{A}X^n} = \frac{1}{2n-1}. \tag{12}$$

Imposing the SSEE algebraic constraint $w_0 = -T_r/M_v$ uniquely fixes the kinetic exponent without any fitting:

$$\boxed{n = \frac{T_r - M_v}{2\,T_r} = \frac{1+w_0}{2w_0} \approx -0.09527,} \tag{13}$$

where $T_r = 3(\varphi+\beta)$ and $M_v = \varphi + \pi + K_v$ are the SSEE structural constants defined in Section 1.2. One verifies immediately that $1/(2n-1) = -T_r/M_v = w_0$ identically.

The overall normalisation is fixed by requiring that the present-day dark-energy density equals the SSEE prediction:

$$\rho_{\mathrm{DE},0} = \frac{3H_0^2}{8\pi G}\,\frac{T_r}{M_v}, \tag{14}$$

with H_0 constrained observationally to $66.75^{+0.44}_{-0.44}$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ by the MCMC analysis of Paper 2. Given the energy scale $\rho_{\mathrm{DE},0}$, the amplitude reads

$$\mathcal{A} = \frac{\rho_{\mathrm{DE},0}^{1-n}}{2n-1}, \tag{15}$$

which completes the specification of $P(X)$ with no free parameters beyond H_0.

Remark on the CPL evolution. The pure k-essence sector (12) delivers a *constant* $w = w_0$. The time variation w_a arises from the viscous source (Section 2.3), and is shown there to emerge algebraically from the scale-factor evolution of the viscosity coefficient.

2.3 Geometric Bulk Viscosity and the Emergence of w_a

2.3.1 Constant-coupling regime (w_0)

The retention constant $\mathrm{KAL}_0 \equiv \beta + \pi \approx 5.521$ governs the dissipative sector. Following the Eckart first-order formalism, the bulk viscosity coefficient is coupled to the instantaneous expansion rate:

$$\zeta_0 = \frac{\mathrm{KAL}_0}{8\pi G}\,H. \tag{16}$$

This delivers a *constant-coupling* viscous pressure $\Pi_0 = -3\zeta_0 H = -3\,\mathrm{KAL}_0 H^2/8\pi G$, which governs the background acceleration (Eq. 28) and fixes the present-day effective equation of state $w_0 = -T_r/M_v$ algebraically (Section 2.2).

Physical motivation for $\zeta \propto H$. Bulk viscosity proportional to the Hubble rate arises naturally in kinetic theory when the dominant relaxation time is set by the expansion itself [Weinberg, 1971, Brevik et al., 2005]. Within SSEE the coupling constant is not fitted but predicted geometrically as $\mathrm{KAL}_0 = (\pi+\varphi)/2 + \pi$.

2.3.2 Scale-factor evolution and the algebraic determination of w_a

The constant-coupling viscosity of the preceding section fixes w_0 algebraically but predicts a *time-independent* effective equation of state. DESI DR2 data require a time-varying component; within SSEE this must arise from a scale-factor-dependent extension of the viscosity coefficient.

Structural requirements on $\zeta(a)$. Any admissible SSEE viscosity extension $\zeta(a)$ must satisfy three independent constraints:

(i) **De Sitter endpoint:** $\zeta(a) \to 0$ as $a \to 1$, so that no viscous term shifts the present-day equation of state away from the k-essence value $w_0 = -T_r/M_v$.

(ii) **Dimensional consistency:** $[\zeta] = [\rho_{\rm DE}/H]$, constructed from the available dynamical quantities $\{\rho_{\rm DE}, H, a\}$.

(iii) **Algebraic closure:** the dimensionless amplitude must be a ratio of SSEE structural constants derived exclusively from $\{\varphi, \pi\}$.

The unique admissible form satisfying (i) and (ii) at leading order in $(1-a)$ — the natural expansion around the de Sitter endpoint — is

$$\zeta(a) = \Gamma \, \frac{(1-a)\,\rho_{\rm DE}(a)}{H(a)}, \tag{17}$$

where Γ is a dimensionless amplitude to be fixed by constraint (iii).

Algebraic determination of Γ. The k-essence sector has already employed the ratio T_r/M_v to fix $|w_0|$. The only independent dimensionless ratio remaining among the SSEE structural constants is P_{sc}/K_v, where $P_{sc} = 2\varphi+\pi \approx 6.378$ (Dynamical Evolution Scalar) and $K_v = 2(\varphi+\pi) \approx 9.519$ (Structural Constraint). The sign is determined by the physical requirement that dark-energy density decreases with time in the observationally favoured phantom-crossing regime:

$$\Gamma = -\frac{P_{sc}}{K_v} = -\frac{2\varphi+\pi}{2(\varphi+\pi)}, \tag{18}$$

yielding the unique SSEE-admissible viscosity:

$$\boxed{\zeta(a) = -\frac{2\varphi+\pi}{2(\varphi+\pi)} \, \frac{(1-a)\,\rho_{\rm DE}(a)}{H}.} \tag{19}$$

Algebraic consistency with CPL. The CPL parametrisation $w(a) = w_0 + w_a(1-a)$ is the standard model-independent description of slowly-varying dark energy; it serves here as the observational language, not as a physical assumption of SSEE. Adopting CPL and substituting $\zeta(a)$ from Eq. (19) into the SSEE conservation equation (29), one finds algebraic consistency if and only if

$$\boxed{w_a = -\frac{P_{sc}}{K_v} = -\frac{2\varphi+\pi}{2(\varphi+\pi)} \approx -0.6699.} \tag{20}$$

This is not a derived prediction in the sense that CPL is deduced from the action; rather, it establishes that *the SSEE viscosity uniquely fixes the CPL coefficient w_a from the algebraic structure of* $\{\varphi, \pi\}$. The quantity w_a is therefore not a free parameter of the model but a computable consequence of the two-axiom system, in the same sense that w_0 is fixed by the k-essence exponent n. At $a = 1$ the viscous correction vanishes ($\zeta \to 0$), recovering the pure k-essence regime $w = w_0$.

The implied dark-energy density evolution reads:

$$\rho_{\rm DE}(a) = \rho_{\rm DE,0}\, a^{-3(1+w_0+w_a)}\, e^{-3w_a(1-a)}, \tag{21}$$

consistent with the CPL conservation equation

$$\dot{\rho}_{\rm DE} + 3H\, \rho_{\rm DE}\,(1 + w_0) = -3H\, w_a\,(1-a)\, \rho_{\rm DE}, \tag{22}$$

with the effective viscous pressure

$$\Pi(a) = w_a\,(1-a)\, \rho_{\rm DE}(a). \tag{23}$$

At $a \ll 1$ the factor $(1 - a) \to 1$ and $\rho_{\rm DE}$ redshifts according to Eq. (21), so ζ remains finite and well-behaved throughout cosmic history.

Boundary conditions. The full viscosity interpolates between two algebraically-fixed regimes:

$$a \to 0\ (\text{early}): \quad \zeta \to -\frac{2\varphi + \pi}{2(\varphi + \pi)}\, \frac{\rho_{\rm DE,0}\, a^{-3(1+w_0+w_a)}\, e^{-3w_a}}{H(a)}, \tag{24}$$

$$a \to 1\ (\text{today}): \quad \zeta \to 0. \tag{25}$$

Both limits are free of divergences and require no new parameters.

The viscous pressure is therefore

$$\Pi(a) = \underbrace{-\frac{3\,\mathrm{KAL}_0\, H^2}{8\pi G}}_{\text{background}} \underbrace{-\ 3H\, w_a\,(1-a)\, \rho_{\rm DE}(a)}_{\text{CPL evolution}}, \tag{26}$$

where the first term is the constant-coupling Eckart pressure that sustains $\ddot{a} > 0$ and the second term generates the running $w(a)$ observed by DESI DR2 [Abdul-Karim et al., 2025].

2.4 Modified Friedmann Equations

Applying the principle of least action ($\delta S = 0$) to the FLRW metric $ds^2 = -dt^2 + a^2(t)\, d\mathbf{x}^2$ yields three governing equations for the SSEE background.

I. Energy equation:

$$H^2 = \frac{8\pi G}{3}\big(\rho_m + \rho_r + \rho_{\rm SSEE}\big). \tag{27}$$

II. Acceleration equation:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left[\rho_m + 2\rho_r + \rho_{\rm SSEE}(1 + 3w_0) - \frac{9\,\mathrm{KAL}_0\, H^2}{8\pi G}\right]. \tag{28}$$

III. Conservation equation with dissipative source:

$$\dot{\rho}_{\rm SSEE} + 3H\,\rho_{\rm SSEE}(1+w_0) = \frac{9\,{\rm KAL}_0\,H^3}{8\pi G}. \tag{29}$$

Derivation of the coefficient 9.

From Eq. (16), the viscous pressure is $\Pi = -3{\rm KAL}_0 H^2/8\pi G$. In the Raychaudhuri equation the effective pressure of the SSEE sector enters as $p_{\rm eff} = w_0\rho_{\rm SSEE} + \Pi$, so the acceleration term becomes

$$\begin{aligned} -\frac{4\pi G}{3}\left(\rho_{\rm SSEE} + 3p_{\rm eff}\right) &= -\frac{4\pi G}{3}\left[\rho_{\rm SSEE}(1+3w_0) + 3\Pi\right] \\ &= -\frac{4\pi G}{3}\left[\rho_{\rm SSEE}(1+3w_0) - \frac{9\,{\rm KAL}_0\,H^2}{8\pi G}\right], \end{aligned} \tag{30}$$

which is exactly Eq. (28). The factor 9 arises from $3 \times |\Pi|/(H^2) = 3 \times 3{\rm KAL}_0/8\pi G$; a factor of 3 rather than 9 would correspond to including Π only once instead of as the trace of the viscous stress.

Similarly, the conservation law $\nabla_\mu T^{\mu\nu} = 0$ for the effective fluid (w_0, Π) gives $\dot{\rho} + 3H(\rho + w_0\rho + \Pi) = 0$, hence $\dot{\rho} + 3H\rho(1+w_0) = 9{\rm KAL}_0 H^3/8\pi G$, confirming that Eq. (28) and Eq. (29) are mutually consistent.

2.5 Israel–Stewart Viscous Perturbations

The Eckart formulation used in §2.2–2.4 is a first-order theory that violates causality [Hiscock & Lindblom, 1985]: the viscous signal propagates instantaneously. Israel–Stewart (IS) theory [Israel, 1976, Israel & Stewart, 1979, Maartens, 1996] resolves this by promoting Π to a dynamical variable with a finite relaxation time τ_Π:

$$\tau_\Pi H_0\,\tilde{h}(a)\,\frac{{\rm d}\tilde{\Pi}}{{\rm d}\ln a} + \tilde{\Pi} = -\,{\rm KAL}_0\,\tilde{h}^2(a), \tag{31}$$

where $\tilde{h} = H/H_0$, $\tilde{\Pi} = \Pi\,8\pi G/H_0^2$, and all densities are normalised to $\rho_{\rm crit,0}$. In the limit $\tau_\Pi \to 0$, Eq. (31) reduces to the Eckart result $\tilde{\Pi} = -{\rm KAL}_0\tilde{h}^2$.

Causality constraint. The bulk-viscous sound speed in IS theory is [Hiscock & Lindblom, 1985]

$$c_s^2 = \frac{\zeta}{\rho_{\rm DE}\,\tau_\Pi} = \frac{{\rm KAL}_0}{3\,\Omega_{\rm DE}\,(\tau_\Pi H_0)} \le 1. \tag{32}$$

Setting $c_s^2 = 1$ (the causality boundary) uniquely fixes the IS relaxation time as an algebraic combination of SSEE constants:

$$\tau_\Pi H_0 \;=\; \frac{{\rm KAL}_0}{3\,\Omega_{\rm DE}} \;=\; \frac{{\rm KAL}_0\,M_v}{3\,T_r} \;\approx\; 2.191, \tag{33}$$

with no free parameters.

Growth index. Coupling Eq. (31) to the conservation equation (29) and the linear-growth ODE

$$\delta'' + \left[2 + \frac{d\ln\tilde{h}}{d\ln a}\right]\delta' = \frac{3}{2}\,\frac{\Omega_{m,\mathrm{dyn}}\,a^{-3}}{\tilde{h}^2(a)}\,\delta, \tag{34}$$

we solve the system numerically (`ssee_is_growth.py`) with the IS background $\tilde{h}^2(a) = \Omega_{m,\mathrm{dyn}}\,a^{-3} + \rho_{\mathrm{DE}}(a)$. The growth rate $f \equiv d\ln\delta/d\ln a \approx \Omega_m(a)^\gamma$ is well described by a power law over $0.1 \le a \le 1$ with

$$\gamma_{\mathrm{bg}} = 0.657 \pm 0.002 \quad \text{(essentially independent of } \tau_\Pi H_0\text{)}. \tag{35}$$

This is the CDM growth index on the IS background, i.e. Eq. (34) without viscous perturbation feedback in the δ equation. It differs by 6% from the algebraic prediction $\gamma = \varphi^{-1} = 0.618$. Paper 3 §5.4 reports both values: $\gamma_{\mathrm{alg}} = 0.618$ and $\gamma_{\mathrm{bg}} = 0.657$. The full IS-coupled perturbation theory of Paper 5, which includes viscous feedback in the growth equation, yields $\gamma_{\mathrm{IS}} = 0.554 \pm 0.001$ — consistent with $\gamma_{\Lambda\mathrm{CDM}} \approx 0.55$ and lower than the background-only estimate here. Whether $\gamma = \varphi^{-1}$ is the true attractor of the IS boundary conditions remains a target prediction for Stage-IV weak-lensing surveys (LSST, Euclid). Note that the direct IS background growth-factor integral (this Appendix) yields $S_8 = 0.837$, while Paper 3's CAMB post-processing method with $\gamma_{\mathrm{bg}} = 0.657$ gives $S_8 = 0.818$; both are consistent with Planck 2018 (0.832 ± 0.013) within 1σ. Paper 5's full IS-coupled perturbation theory (including viscous feedback in δ), with $\Omega_{m,\mathrm{cosm}} = 0.320$ in the Poisson source, yields $\sigma_8 = 0.820$ and $S_8 = 0.847$ (3.9σ KiDS, an open tension) — the more rigorous estimate.

S_8 constraint. Using the IS growth factor $D_{\mathrm{IS}}(a)$ and the CMB-sector matter density $\Omega_{m,\mathrm{CMB}} = \Omega_{m,\mathrm{dyn}} \times \mathcal{M} = 0.3199$ (MIRA, Paper 3),

$$S_8^{\mathrm{IS}} \equiv \sigma_{8,\mathrm{IS}}\sqrt{\frac{\Omega_{m,\mathrm{CMB}}}{0.3}} \approx 0.837, \tag{36}$$

within 0.4σ of the Planck-2018 value 0.832 ± 0.013 [Planck Collaboration VI, 2020]. Using the dynamic sector $\Omega_{m,\mathrm{dyn}} = 0.160$ instead gives $S_8 \approx 0.59$ (6σ tension); the correct physical choice depends on the formal resolution of the two-sector Friedmann problem (Section 2.8, item 3).

2.6 Physical Regimes

Late universe ($z \lesssim 2$, DESI DR2). At low redshift $H \approx H_0$, so the viscous source $\propto \mathrm{KAL}_0 H^3/8\pi G$ is small compared to ρ_{SSEE}. The acceleration equation is dominated by $\rho_{\mathrm{SSEE}}(1+3w_0) < 0$ (since $w_0 \approx -0.840 < -1/3$), driving $\ddot{a} > 0$ without dark matter. This regime is the one tested against DESI DR2 in Paper 2, where the algebraic prediction (w_0, w_a) matches the data at 0.05σ.

Early universe (CMB epoch, $z \sim 10^3$). At recombination $H \gg H_0$, the geometric retention term $9\,\mathrm{KAL}_0 H^3/8\pi G$ acts as a dissipative source in Eq. (29), injecting energy into the dark-energy fluid and modifying its effective equation of state. This structural viscosity provides the physical mechanism by which an effective matter density $\Omega_{m,\mathrm{eff}} = 1 - T_r/M_v \approx 0.160$ yields the correct CMB acoustic scale through the MIRA factor $\mathcal{M} = (3\varphi + \pi)/4 \approx 1.9989$ (Paper 3, §2.3).

2.7 Inflationary Embedding: α-Attractor with $\alpha = \varphi^4/3$

The SSEE algebraic predictions for the primordial scalar spectral index $n_s = 1 - \varphi^{-7}$ (Paper 1, §1.1) and the tensor-to-scalar ratio $r = \varphi^{-10}$ (Paper 4) belong simultaneously to a well-defined class of inflationary models [Guth, 1981, Linde, 1982]: the α-*attractors* of Kallosh & Linde [2013].

α-attractor predictions. In the universal leading-order limit, α-attractor models predict

$$n_s = 1 - \frac{2}{N}, \qquad r = \frac{12\alpha}{N^2}, \tag{37}$$

where N is the number of inflationary e-folds and $\alpha > 0$ parameterises the inverse curvature of the scalar field's Kähler target manifold [Kallosh & Linde, 2013, Ferrara et al., 2013]. Setting $\alpha = 1$ recovers Starobinsky R^2 inflation [Starobinsky, 1980] exactly.

SSEE consistency check. The SSEE axioms pre-commit two *independent* inflationary observables: $n_s = 1-\varphi^{-7}$ (Axiom 1, spectral tilt) and $r = \varphi^{-10}$ (Axiom 2, tensor-to-scalar ratio). Neither is derived from the α-attractor framework; they are algebraic predictions from $\{\varphi, \pi\}$ that can be tested for *mutual consistency* within that framework.

From Axiom 1 alone, the standard slow-roll relation $n_s = 1 - 2/N$ gives

$$N = \frac{2}{\varphi^{-7}} = 2\varphi^7 \approx 58.07, \tag{38}$$

which lies squarely in the observationally required window $50 \lesssim N \lesssim 60$ [Planck Collaboration X, 2020].

Independently, inserting $N = 2\varphi^7$ and Axiom 2 ($r = \varphi^{-10}$) into the second attractor relation $r = 12\alpha/N^2$:

$$\alpha = \frac{r\,N^2}{12} = \frac{\varphi^{-10}\cdot 4\varphi^{14}}{12} = \frac{4\varphi^4}{12} = \frac{\varphi^4}{3}. \tag{39}$$

This is a *non-trivial internal consistency*: two independently pre-committed predictions simultaneously select a unique α-attractor geometry. The identification is algebraically exact (verified to machine precision; see `ssee_inflation_connection.py`). Using the Fibonacci identity $\varphi^4 = 3\varphi + 2$:

$$\alpha = \frac{3\varphi + 2}{3} = \varphi + \frac{2}{3} \approx 2.285. \tag{40}$$

Geometric interpretation. In the α-attractor framework the Kähler curvature of the inflaton's target space (a Poincaré disk of radius $\sqrt{3\alpha}$) is $\mathcal{R} = -2/(3\alpha)$. For $\alpha = \varphi^4/3$:

$$\mathcal{R} = -\frac{2}{3\cdot\varphi^4/3} = -\frac{2}{\varphi^4} = -2\varphi^{-4} \approx -0.292. \tag{41}$$

The inflaton therefore lives on a hyperbolic manifold whose curvature is set by φ alone. This gives a *geometric* origin for the appearance of φ in inflationary observables: φ enters not as a numerical coincidence but as the intrinsic length scale of the Kähler geometry of the SSEE inflaton sector.

Slow-roll parameters. The SSEE predictions $(n_s, r) = (1-\varphi^{-7}, \varphi^{-10})$ fix the slow-roll parameters uniquely:

$$\varepsilon = \frac{r}{16} = \frac{\varphi^{-10}}{16}, \tag{42}$$

$$\eta = \frac{n_s - 1 + 6\varepsilon}{2} = \frac{-\varphi^{-7} + 3\varphi^{-10}/8}{2}, \tag{43}$$

$$\frac{\eta}{\varepsilon} = 3 - 8\varphi^3 = -5 - 16\varphi \approx -30.9. \tag{44}$$

The ratio $\eta/\varepsilon = 3-8\varphi^3$ is a pure algebraic identity derived from the SSEE axioms $\{\varphi, \pi\}$; it defines a one-parameter family of inflationary potentials $V(\phi_{\mathrm{inf}})$ whose single element consistent with α-attractors is $\alpha = \varphi^4/3$.

Relation to Starobinsky. Starobinsky ($\alpha = 1$) reproduces n_s correctly for $N = 2\varphi^7$ but gives $r_{\mathrm{Staro}} = 3/\varphi^{14} \approx 0.00356$, a factor of $\varphi^4/3 \approx 2.28$ below the SSEE prediction φ^{-10}. SSEE is therefore *not* Starobinsky inflation; it is a generalised α-attractor that reduces to Starobinsky only in the limit $\varphi \to 1$ (i.e. $\alpha \to 1$, the fixed point of the golden-ratio recursion $x = 1 + 1/x$).

2.8 Summary and Open Problems

Table 2 summarises the EFT status. The action (10) with kinetic exponent (13) and viscosity (19) constitutes a self-consistent, zero-free-parameter description of the SSEE background evolution. Every constant (n, KAL_0, w_0, w_a, Ω_{DE}) is derived from φ and π alone; only the overall energy scale H_0 enters as an observational input. Notably, $w_a = -P_{sc}/K_v$ is *not* postulated but emerges as the unique algebraic amplitude of the viscosity's scale-factor evolution (Section 2.3.2). The gradient instability of the power-law IR sector ($c_s^2 < 0$) is resolved by the canonical replacement $K(X) = X/\mathrm{KAL}_0$ (Papers 7 and 10); the present section focuses on the background algebraic structure.

Four aspects require further development:

1. **Growth index.** The background IS growth index $\gamma_{\mathrm{bg}} = 0.657$ (§2.5) differs by 6% from the algebraic prediction $\varphi^{-1} = 0.618$. The full IS-coupled result (Paper 5) gives $\gamma_{\mathrm{IS}} = 0.554$, closer to ΛCDM. A mechanism selecting $\gamma = \varphi^{-1}$ from IS boundary conditions is not yet identified; this constitutes a near-prediction testable by LSST/Euclid Stage-IV weak lensing.

2. **Full perturbation spectrum.** The IS growth ODE (34) governs sub-Hubble dark-matter perturbations [Mukhanov et al., 1992]; the matching to CMB scalar transfer functions and the tensor-mode propagation in the IS background require a complete Boltzmann implementation.

3. **Radiative stability.** The EFT cutoff scale Λ_{UV} and the quantum stability of the $n < 0$ power-law sector are not yet established.

4. **Inflation–dark energy unification.** The α-attractor sector (§2.7) and the k-essence dark-energy sector (§2.2) both originate from φ, suggesting a single unified action. Constructing the quintessential inflation potential $V(\phi)$ that reduces to $P(X) = \mathcal{A}X^n$ at late times and to an $\alpha = \varphi^4/3$ attractor at early times is identified as high-priority future work.

Table 2: EFT status summary for SSEE-V3.6.

Element	Status	Derivation
Action form S	✓Complete	Standard GR + k-essence + Eckart
$u_\mu(\phi)$ coupling	✓Complete	$u_\mu = -\partial_\mu\phi/\sqrt{2X}$
Kinetic exponent n	✓Algebraic	$n = (T_r - M_v)/2T_r$ from φ, π
Equation of state w_0	✓Algebraic	$w_0 = 1/(2n-1) = -T_r/M_v$
Viscosity coupling ζ	✓Algebraic	$\zeta = \mathrm{KAL}_0 H/8\pi G$
Friedmann eqs. I–III	✓Consistent	Factor 9 verified in §2.4
w_a from EFT	✓Algebraic	$\zeta(a) \propto (1-a)\rho_{\mathrm{DE}}/H$; see §2.3.2
IS relaxation time τ_Π	✓Algebraic	$\tau_\Pi H_0 = \mathrm{KAL}_0/(3\Omega_{\mathrm{DE}}) \approx 2.191$
Gradient stability	∘ IR ansatz unstable	Power-law $P(X) \propto X^n$: $c_s^2 \approx -0.84$ (resolved in
$K(X) = X/\mathrm{KAL}_0$ stability	✓Verified	Papers 7–10: $c_s^2 = 1$ (canonical); $c_s^2 \in (1/3, 1)$ (U
Growth index γ_{bg}	✓Numerical	0.657 ± 0.002 (IS background, CDM growth eq
S_8 (CMB sector)	✓Numerical	0.837 (IS bkg); 0.847 (full IS, Paper 5; 3.9σ Ki
α-attractor ($\alpha = \varphi^4/3$)	✓Algebraic	Eq. (39); exact, $N = 2\varphi^7$
Kähler curvature $\mathcal{R} = -2\varphi^{-4}$	✓Algebraic	Eq. (41)
Full perturbation spectrum	∘ Open	Off-diagonal CMB cov. + Stage-IV WL
Inflation–DE unification	∘ Open	Quintessential inflation potential
Radiative stability	∘ Open	Future work

3 Dark Energy Evolution

SSEE models dynamical dark energy via the Chevallier–Polarski–Linder (CPL) form [Chevallier & Polarski, 2001, Linder, 2003] $w(a) = w_0 + w_a(1-a)$. Motivated by Eq. (9), the expansion is subject to an effective geometric pressure:

$$H^2 = \frac{8\pi G}{3}(\rho_{\mathrm{bar}} + \rho_{\mathrm{DE}}), \qquad \rho_{\mathrm{DE}} = \rho_{\mathrm{crit},0}\left(\frac{T_r}{M_v}\right). \tag{45}$$

The baseline state w_0 is set by the static geometric ratio T_r/M_v, and the evolution w_a encodes the dynamical departure from saturation driven by P/K_v:

$$w_0 = -\frac{T_r}{M_v} \approx -0.840, \qquad w_a = -\frac{P}{K_v} \approx -0.670. \tag{46}$$

Explicit reduction to the two axioms. Substituting the algebraic definitions $T_r = \frac{3(3\varphi+\pi)}{2}$ (TRIAL), $M_v = 3(\varphi+\pi)$ (MIKAEL_V), $P_{sc} = 2\varphi+\pi$ (PYROS), and $K_v = 2(\varphi+\pi)$ (KRYSTOS), both dark-energy observables collapse to closed forms in φ and π alone:

$$\boxed{w_0 = -\frac{3\varphi+\pi}{2(\varphi+\pi)} = -0.8399}, \qquad \boxed{w_a = -\frac{2\varphi+\pi}{2(\varphi+\pi)} = -0.6699}. \tag{47}$$

No fitting procedure is involved: these values are uniquely fixed by the geometry of φ and π. The numerators $3\varphi + \pi$ and $2\varphi + \pi$ reflect the three- and two-fold contributions of φ to the saturation horizon, while the common denominator $2(\varphi + \pi)$ is twice the Stability Metric $\Omega = \varphi + \pi$.

Asymptotic saturation and phantom crossing. The values w_0 and w_a are derived as Taylor coefficients of the full effective equation of state around $a = 1$. While the CPL parametrisation yields $w_0 + w_a = -1.510$, we employ it strictly within the low-redshift domain of validity ($z \leq 2.3$) for observational comparisons, noting that asymptotic stability at $a \to \infty$ requires the full non-linear saturation boundary T_r/M_v rather than the linear CPL extrapolation.

The CPL form also predicts a *phantom crossing*: setting $w(a_*) = -1$ gives

$$a_* = 1 + \frac{1 + w_0}{w_a} = 1 + \frac{1 + (-0.840)}{-0.670} \approx 0.761 \quad (z_* \approx 0.31). \tag{48}$$

For $a < a_*$ (i.e. $z > 0.31$), the effective equation of state $w < -1$ (phantom-like [Caldwell, 2002]); for $a > a_*$, $w > -1$ (quintessence-like), reaching $w_0 = -0.840$ today. This behaviour is inherent to the linear CPL approximation and does not signal a ghost or gradient instability in the SSEE action. However, the effective CPL equation of state implies NEC violation ($\rho + p < 0$, equivalently $1 + w < 0$) for all $z > 0.31$ at the background level. In the full SSEE action, the viscous pressure Π partially compensates this violation at the perturbation level; a complete NEC analysis at the level of the Israel–Stewart stress tensor [Israel, 1976, Israel & Stewart, 1979] is deferred to Paper 4 (see App. A, EFT embedding).

DESI DR2 [Abdul-Karim et al., 2025] analyses indicate an evolving preference for dynamical dark energy. The SSEE algebraic point $(w_0, w_a) = (-0.840, -0.670)$ falls within the 68% confidence contours of the combined BAO+CMB+DESY5 datasets (Mahalanobis $\chi^2_{2D} = 0.080$, i.e. 0.05σ; quantified in Paper 2 [Almeida Vallejo, 2026]).

4 Galaxy Cluster Dynamics

4.1 The Cluster Mass Discrepancy and IGIMF

In rich galaxy clusters, inferred dynamical mass typically exceeds the visible baryonic mass by a factor of ∼5–6. Recent literature incorporating a variable integrated galaxy-wide initial mass function (IGIMF) [Kroupa & Weidner, 2003, Zhang et al., 2026] demonstrates that baryonic masses have been systematically underestimated: heavy stellar remnants raise the effective baryonic baseline to at least $88^{+5}_{-4}\%$ of the MOND dynamical mass. In SSEE, this updated IGIMF baseline multiplied by $\mathrm{KAL}_0 \approx 5.5214$ natively closes the remaining discrepancy without cold dark matter.

4.2 Structural Viscosity and the Neutrino Bridge

At the cluster scale, the total effective dynamical mass M_{dyn} within radius R is:

$$M_{\mathrm{dyn}}(R) = M_{\mathrm{bar}}^{\mathrm{IGIMF}}(R) \cdot \mathrm{KAL}_0 \cdot [1 + f_\nu], \tag{49}$$

where f_ν quantifies the fractional mass contribution of trapped relic neutrinos. Using $\sum m_\nu^{\mathrm{active}} = 0.0824\,\mathrm{eV}$ (the SSEE-derived value, Paper 4) the cosmological neutrino density is $\rho_\nu \approx 27.7\,\mathrm{eV\,cm^{-3}}$. For a cluster with escape velocity $v_{\mathrm{esc}} \approx 2000\,\mathrm{km\,s^{-1}}$, integration of the Fermi–Dirac distribution [Tremaine & Gunn, 1979] yields a trapped fraction $f_\nu \approx 0.018$–0.022. This structurally consistent ∼2% correction provides the final macroscopic bridge.

4.3 Quantitative Cluster Analysis

Applying Eq. (49) to four well-documented clusters demonstrates predictive accuracy against observed dynamical masses (Table 3).

Table 3: Cluster mass predictions. All masses within R_{200} in units of $10^{14}\,M_\odot$. $M_{\rm SSEE} = M_{\rm bar}^{\rm IGIMF} \times 5.5214 \times 1.02$. IGIMF baselines and observed masses from Zhang et al. [2026]; Bullet lensing mass from Clowe et al. 2006.

Cluster	$M_{\rm bar}^{\rm IGIMF}$	$M_{\rm SSEE}$ $[10^{14}\,M_\odot]$	$M_{\rm obs}$
Coma	1.8 ± 0.2	10.1 ± 1.1	9.8 ± 1.0
A2029	2.2 ± 0.2	12.4 ± 1.1	12.0 ± 1.2
A478	1.5 ± 0.2	8.4 ± 1.1	8.0 ± 1.0
Bullet (main)	1.2 ± 0.2	6.7 ± 1.1	6.5 ± 1.0

All four predictions agree with observed masses at the 1σ level ($\chi_r^2 = 0.122$; full analysis in Paper 2 [Almeida Vallejo, 2026]).

4.4 Gravitational Lensing and the Newtonian Limit

The Bullet Cluster (1E 0657-56) is frequently cited as proof of particulate dark matter due to the spatial offset between collisional X-ray gas and lensing-mass peaks [Clowe et al., 2006]. SSEE proposes a qualitative mechanism for this offset: the environment-dependent amplifier $\mathrm{KAL}(x)$ (Eq. 50) selectively enhances the effective surface mass density in compact galactic potential wells (high x) relative to the diffuse stripped gas (low x), displacing lensing peaks toward the galaxies without collisionless particles. A quantitative calculation of the projected convergence $\kappa(\vec{\theta})$ from first principles is deferred to future work.

The structural viscosity acts as an environment-dependent scalar field. Defining $x = |\nabla\Phi|/a_0$, SSEE adopts an interpolation function analogous to the canonical MOND [Milgrom, 1983, Famaey & McGaugh, 2012] simple interpolating function $\mu(x)$:

$$\mu_{\rm SSEE}(x) = \frac{x+1}{x+\mathrm{KAL}_0} \implies \mathrm{KAL}(x) = \frac{1}{\mu_{\rm SSEE}(x)} = \frac{x+\mathrm{KAL}_0}{x+1}. \tag{50}$$

For $x \gg 1$ (Newtonian limit), $\mathrm{KAL}(x) \to 1$. For $x \ll 1$ (deep MONDian limit), $\mathrm{KAL}(x) \to \mathrm{KAL}_0$. During a high-velocity cluster collision, the stripped X-ray gas flattens its local potential gradient ($x \ll 1$), activating the full KAL_0 amplification; collisionless galaxies retain compact potential wells ($x \geq 1$), coupling non-linearly.

The effective lensing convergence is:

$$\kappa(\vec{\theta}) = \frac{\Sigma_{\rm SSEE}(\vec{\theta})}{\Sigma_{\rm crit}}, \quad \Sigma_{\rm SSEE}(\vec{\theta}) = \int \rho_{\rm bar}^{\rm IGIMF}(\vec{\theta}, \ell) \cdot \mathrm{KAL}(x)\, d\ell, \quad \Sigma_{\rm crit} = \frac{c^2}{4\pi G}\frac{D_S}{D_L D_{LS}}. \tag{51}$$

This allows lensing peaks to track galaxies, consistent with QUMOND simulations [Angus et al., 2007].

Open problem: dimensional connection to fundamental scales. The algebraic prediction $H_0 = 3(\varphi+\pi)^2 \approx 67.96\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ is a numerical coincidence (see Appendix A, footnote †): the adimensional combination $3(\varphi+\pi)^2$ matches the Hubble constant in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ units, but no theoretical bridge connects $\{\varphi, \pi\}$ to the ratio of kilometre, second, and megaparsec scales. A complete first-principles derivation would require a quantum-gravity argument linking the Planck length ($\ell_{\rm Pl} \approx 1.6 \times 10^{-35}$ m) to the cosmic expansion rate ($1\,\mathrm{Mpc} \approx 3.1 \times 10^{22}$ m), a connection 57 orders of magnitude in scale. This derivation lies outside the scope of the present framework and is identified as an open problem for future theoretical work; it is the reason H_0 is taken as the single dimensional anchor of Postulate D (§1.3) rather than a derived quantity.

Data Availability

All observational data used in this work are publicly available. DESI DR2 BAO measurements are from Abdul-Karim et al. [2025] (https://arxiv.org/abs/2503.14738). Galaxy-cluster dynamical masses are from Zhang et al. [2026] (https://arxiv.org/abs/2602.06082). Analysis scripts and data files are available at https://github.com/mikealmeida1721/SSEE.

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5 Domain of Validity

A persistent risk in multi-sector theoretical models is that statements of validity or falsification are conflated across domains that probe different physical regimes. This section explicitly delineates the domain of validity of each SSEE sector and the dataset used to validate it. This delineation applies uniformly to Papers 1–4.

Dynamic sector (low redshift, $z \lesssim 2.3$). Governed by $\Omega_{m,\mathrm{dyn}} = 0.1601$, $w_0 = -0.8399$, $w_a = -0.6699$. Validated against: DESI DR2 BAO (13 measurements, $0.295 \leq z \leq 2.33$), galaxy-cluster mass data (IGIMF baseline, Paper 2). *Result*: $\Delta\mathrm{BIC} = -5.55$ (dynamic sector, Paper 2), 0.05σ from DESI DR2 CPL posterior in the w_0–w_a plane. *Known limitation*: under standard Friedmann background, $\chi^2_r(H(z)) = 1.861$ (Cosmic Chronometers); this is a direct consequence of the enlarged $r_d = 175.6$ Mpc and is documented, not hidden (Paper 2 §5.3).

Observational sector (high redshift, CMB). Governed by $\Omega_{m,\mathrm{CMB}} = \Omega_{m,\mathrm{dyn}} \times \mathrm{MIRA} = 0.3199$, where $\mathrm{MIRA} = (3\varphi + \pi)/4 = 1.99892$ is algebraically pre-committed (January 2026; Genesis 5.12, Zenodo 10.5281/zenodo.19679049). Validated against: Planck PR4 TT+TE+EE+lensing via CAMB; `plik_lite` TT-TEEE likelihood via Cobaya (Paper 3). *Result*: $\chi^2_r = 1.062$ (TT, 1971 multipoles); $\Delta\mathrm{BIC} = -31.3$ at $H_0 = 67.075\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ ($k = 2$ vs $k = 6$ for ΛCDM; canonical conservative estimate from Paper 3). Aggressive lower bound with $k = 1$: $\Delta\mathrm{BIC} = -40.3$. *Claim*: SSEE is *not falsified* by the Planck PR4 CMB spectra. This claim is sector-specific: it refers to the observational sector (with MIRA applied), not the dynamic sector directly.

Algebraic sector (purely geometric). All constants fixed from $\{\varphi, \pi\}$ with zero data input. Cross-validated against: Planck 2018 (n_s, $\Omega_b h^2$, H_0), LiteBIRD forecast (r), BBN ($\sum m_\nu$), Paper 4. *Result*: most Type A predictions in Appendix B, Table 6 are within 2σ of published observational values (see Paper 4 for full compilation). The baryon density $\Omega_b h^2 = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.02242$ (0.32σ from Planck 2018, 0.02237 ± 0.00015) is classified as a *retrodiction*: its structural algebraic form was identified after the data, superseding the preliminary placeholder $3(\pi - \varphi)/200$ (documented in Table 6 and `AUDIT.md`). *Claim*: the algebraic sector is *not falsified* by any currently available dataset at the $> 3\sigma$ level. A future falsification would require $n_s \neq 1 - \varphi^{-7}$ at $> 3\sigma$ from Planck PR4, or $r \neq \varphi^{-10}$ from LiteBIRD.

5.1 Unified Model-Comparison Table

Table 4 consolidates every BIC comparison reported across the four manuscripts into a single cross-comparable register. Each row specifies the dataset, N, the cosmological background assumed, k for SSEE and ΛCDM, and the physical hypothesis tested. No BIC result outside this table is claimed as a model-selection outcome.

Table 4: Unified BIC comparison register (all four papers). $\Delta\mathrm{BIC} < 0$: SSEE preferred; > 0: ΛCDM preferred. "Fails" denotes a domain-specific tension, not global falsification.

Dataset	N	Background	k_{SSEE}	k_Λ	ΔBIC	Hypothesis
DESI DR2 + Planck prior	16	Std. Friedmann	2	3	+206	Full model: background incompatibility with ΛCDM-calibrated Ω_m
DESI DR2 + Planck prior	16	Std. Friedmann	1	3	−5.55	Dynamic sector: w_0, w_a fixed, H_0 free
Planck PR4 diagonal	5914	MIRA-corr.	2	6	+31.1	CMB spectra, conservative k upper bound
`plik_lite` TT-TEEE	6469	MIRA-corr.	2	6	**−31.3**	CMB parsimony: canonical conservative ($k = 2$, Paper 3)
`plik_lite` TT-TEEE	6469	MIRA-corr.	1	6	−40.3	CMB parsimony: aggressive lower bound ($k = 1$)

Reading guide. The +206 and −5.55 use the same 16 data points but test different hypotheses: the former penalises background geometry; the latter isolates w_0, w_a. The −31.3 (canonical, $k = 2$) and +31.1 both use CMB data but differ in likelihood type and k; the aggressive $k = 1$ estimate gives −40.3. *No single BIC summarises the entire framework*: each answers a specific physical question stated in the table.

5.2 Falsification Manifesto

Global falsification rule (theory-level).

Global Falsification Criterion

SSEE-V3.6 is falsified **as a whole** if **any single** sector criterion in Table 5 is met at the stated significance threshold. No parameter re-adjustment is possible because all sectors derive from the same two axioms $\{\varphi, \pi\}$: a failure in one sector cannot be repaired without modifying the axioms, which would change the predictions in all other sectors simultaneously.

Corollary: the sector-specific structure (dynamic / observational / algebraic) exists to delineate which datasets test which part of the framework, not to create rescue mechanisms. If DESI DR5+ excludes $w_0 = -0.840$ at $> 3\sigma$, SSEE-V3.6 is rejected; the observational or algebraic sectors do not save it.

To ensure the framework cannot absorb discrepancies via sector-switching, we also state sector-specific criteria below. These apply uniformly across all four papers and cannot be re-interpreted post-hoc.

SSEE Falsification Criteria

Algebraic sector is falsified if:
- $n_s \neq 1 - \varphi^{-7}$ at $> 3\sigma$ from Planck PR4 final; or
- $r \neq \varphi^{-10}$ at $> 2\sigma$ from LiteBIRD (∼2032).

Dynamic sector is falsified if:
- DESI DR5+ posterior excludes $(-0.840, -0.670)$ at $> 3\sigma$; or
- $\chi^2_r(H(z)) > 2.5$ with full systematic covariance and $\Omega_{m,\text{dyn}} = 0.160$.

Observational sector is falsified if:
- Euclid/KiDS/DES require $\Omega_m < 0.25$ at $z < 1$ (violates the MIRA two-sector crossover); or
- $\Omega_{m,\text{CMB}}/\Omega_{m,\text{dyn}}$ deviates from MIRA = 1.998924 at $> 3\sigma$ from joint CMB+BAO analysis.

What is NOT a falsification:
- A ΔBIC > 0 from applying ΛCDM-calibrated priors to the SSEE background (see +206 in Table 4).
- A $\chi^2_r > 1$ under a non-equivalent cosmological background.
- The Ω_m tension between sectors, provided the MIRA bridge is algebraically self-consistent (verified to 9 s.f., Table 6).

Table 5: Falsification summary: one-sentence decision rule per sector. This table is the binding reference for all four papers. "Tension" = quantified mismatch within a defined domain; "Falsification" = theory-level rejection.

Sector	Validated by	Falsified by ($> 3\sigma$)	Not a falsification
Algebraic	Nine Sovereignties closure; cross-domain Type A table	$n_s \neq 1 - \varphi^{-7}$ (Planck PR4) or $r \neq \varphi^{-10}$ (LiteBIRD)	Any tension arising from unit-convention mismatch
Dynamic	DESI DR2 ΔBIC $= -5.55$; w_0, w_a within 0.05σ	DESI DR5+ excludes $(-0.840, -0.670)$; or $\chi^2_r(H(z)) > 2.5$ with full cov.	ΔBIC $= +206$ from ΛCDM-calibrated Ω_m prior
Observational	Planck PR4 ΔBIC $= -31.3$ (canonical, $k = 2$); $r_d^{\mathrm{SSEE+MIRA}} = 147.16$ Mpc (0.25σ)	Euclid/KiDS $\Omega_m < 0.25$ at $z < 1$; or MIRA ratio $\neq 1.999$ at $> 3\sigma$	$\chi^2_r = 1.062$ vs ΛCDM's 1.043 (non-equivalent background)

On cross-sector BIC comparisons. The $\Delta\mathrm{BIC} = -5.55$ and -31.3 are not directly comparable (different N, backgrounds, and ΛCDM k counts). The $\Delta\mathrm{BIC} = +31.1$ and -31.3 (-40.3 aggressive) bracket the CMB result's sensitivity to the assumed k_{SSEE} (see Paper 3 Table 7). The authoritative reference is Table 4.

A Master Derivation Chain: $\varphi, \pi \to$ Observables

Table 6 provides the complete, unambiguous derivation chain from the two axioms (φ, π) to every cosmological observable used in Papers 1–4. This table is the authoritative cross-reference for all four manuscripts. Each intermediate constant is labelled as either a *primary register* (derived in one algebraic step from φ and π) or a *secondary register* (derived from primary registers).

Table 6: Master derivation chain for the SSEEframework. Result types: **A** = pure algebraic (no data input), **M** = MCMC-validated (data-constrained posterior), **P** = prior-dependent (requires external dataset anchor). $\varphi = (1+\sqrt{5})/2 \approx 1.61803$; $\pi \approx 3.14159$.

Symbol	Definition	Value	Observable link	Type
Level 0 — Axioms				
φ	$(1+\sqrt{5})/2$	1.618034	All	A
π	Archimedes constant	3.141593	All	A
Level 1 — Primary Registers				
Ω_{DNAV}	$\varphi+\pi$	4.759627	Stability metric	A
β	$(\varphi+\pi)/2$	2.379814	Base coupling	A
P_{sc}	$\Omega_{\text{DNAV}}+\varphi$	6.377661	Dyn. evolution	A
AURA	$\varphi+\beta$	3.997847	Resonance limit	A
Level 2 — Secondary Registers				
KAL_0	$\beta+\pi$	5.521406	Viscosity; M_{cluster}	A
K_v	$\varphi+\pi+\Omega_{\text{DNAV}}$	9.519254	Struct. constraint	A
T_r	$3(\varphi+\beta)$	11.993520	Saturation horizon	A
MIRA	$\text{AURA}/2 = (\varphi+\beta)/2$	1.998924	CMB–dyn. bridge	A*
M_v	$\varphi+\pi+K_v$	14.278880	Max. dim. invariant	A
Level 3 — Cosmological Observables				
w_0	$-T_r/M_v$	−0.8399	DESI DR2	A
w_a	$-P_{sc}/K_v$	−0.6699	DESI DR2	A
H_0^{alg}	$3(\varphi+\pi)^2$	67.962 km/s/Mpc	Planck 2018	P†
H_0^{MCMC}	CMB full posterior (Paper 3)	67.08 ± 0.03	Planck PR4	M
$\Omega_{m,\text{dyn}}$	$(\pi-\varphi)/[2(\varphi+\pi)]$	0.1601	Dynamic sector; $1+w_0$ exact	A
Ω_m^{CMB}	$(\pi-\varphi)/(\pi+\varphi)$	0.3201§	Planck 2018	A
n_s	$1-\varphi^{-7}$	0.96556	Planck 2018	A
$\Omega_b h^2$	$(\pi-\varphi)/[3(\varphi+\pi)^2]$	0.02242 (0.32σ)	Planck 2018	A‡
$\Omega_c h^2$	$\text{KAL}_0 \times \Omega_b h^2 \times n_s$	0.11951 (0.4σ)	Planck 2018	A
r	φ^{-10}	0.00813	LiteBIRD 2032	A
$\sum m_\nu^{\text{active}}$	$\mathcal{R}\,\Omega_b h^2 \cdot 94.07/(\tau_{\Pi} H_0)$ (Paper 4)	0.0824 eV	Planck+BBN	P

* MIRA $= (\varphi+\beta)/2 = (3\varphi+\pi)/4 = 1.998924\ldots$ is the canonical value used in all four manuscripts. It serves a dual role: (i) as a *derived constant* (algebraically fixed from $\{\varphi, \pi\}$ via $\beta = (\varphi+\pi)/2$, committed January 2026 before any CMB analysis; see Paper 3), and (ii) as a *sector-mapping operator* projecting $\Omega_{m,\text{dyn}} = 0.1601$ onto $\Omega_{m,\text{CMB}} = 0.3199$ via $\Omega_{m,\text{CMB}} = \Omega_{m,\text{dyn}} \times \text{MIRA} = 0.3199$. Note: in early internal documents MIRA was loosely approximated as $\varphi \approx 1.618$; the canonical value is 1.998924 throughout all four papers. Similarly, KAL_0 is both a derived constant (Level 2) and a dynamical viscosity operator scaling cluster masses.

‡ $\Omega_b h^2 = (\pi-\varphi)/[3(\varphi+\pi)^2] = 0.02242$ is at 0.32σ from Planck 2018 (0.02237 ± 0.00015). This is a *retrodiction*: the structural form was identified after the data (OP-1, `OPEN_PROBLEMS.md`), superseding the preliminary placeholder $3(\pi-\varphi)/200 = 0.02285$ (3.2σ), whose factor 200 was not an algebraic $\{\varphi, \pi\}$ quantity. Using the algebraic input $\Omega_b h^2 = 0.02242$, the formula $\text{KAL}_0 \times \Omega_b h^2 \times n_s$ gives $\Omega_c h^2 = 5.5214 \times 0.02242 \times 0.96556 = 0.11951$ (0.41σ from Planck 0.1200 ± 0.0012).

§ The formula $(\pi-\varphi)/(\pi+\varphi)$ gives 0.32010; the canonical value from the derivation chain ($\Omega_{m,\text{dyn}} \times \text{MIRA}$) is 0.31993. Both round to 0.320 at the reported precision; the MIRA formula is the canonical value used throughout Papers 1–4.

† $H_0^{\text{alg}} = 3(\varphi+\pi)^2 \approx 67.962\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ is a **numerical coincidence** (Type P): the adimensional combination $3(\varphi+\pi)^2$ happens to match the Hubble constant in $\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ units. No dimensional derivation linking $\{\varphi, \pi\}$ to the physical energy scale of Hubble expansion has been established; the formula is retained as an observed coincidence and is not used as input to any other result in this series.

B Unified Symbol Glossary

To avoid ambiguity across Papers 1–4, Table 7 provides the canonical definition of every non-standard symbol used in this series.

Table 7: Unified symbol glossary for the SSEEfour-paper series. All symbols are dimensionless unless otherwise stated.

Symbol	Full name	First defined
φ	Golden ratio $(1+\sqrt{5})/2$	Paper 1 §1
Ω_{DNAV}	Dark Navigation scalar $(\varphi+\pi)$	Paper 1 §1
β	Base coupling scalar	Paper 1 §1
AURA	Acoustic Unified Resonance Amplitude $(\varphi+\beta=(3\varphi+\pi)/2\approx 3.998)$	Paper 1 §2
MIRA	Mapped Inference Resonance Amplitude $(\text{AURA}/2=(3\varphi+\pi)/4\approx 1.9989)$	Paper 3 §2.4
KAL_0	Asymptotic structural viscosity factor $(\beta+\pi)$	Paper 1 §1
P_{sc}	Dynamical evolution scalar $(\Omega+\varphi)$	Paper 1 §1
K_v	Structural constraint invariant	Paper 1 §1
T_r	3D saturation horizon $(3(\varphi+\beta))$	Paper 1 §1
M_v	Maximal dimensional invariant $(\varphi+\pi+K_v)$	Paper 1 §1
$\Omega_{m,\text{dyn}}$	Dynamic matter density (SSEE sector)	Paper 2 §2
$\Omega_{m,\text{CMB}}$	CMB-observed matter density $(\text{MIRA}\cdot\Omega_{m,\text{dyn}})$	Paper 3 §2.4
γ_{IS}	Israel–Stewart growth index (0.554 ± 0.001)	Paper 5 §6.1
$\tau_{\Pi}H_0$	IS relaxation time (dimensionless, ≈ 2.191)	Paper 1 App.A

C Result-Type Classification Protocol

A recurring risk in multi-paper series is that results of different epistemic status are compared without adequate labelling. To ensure methodological transparency across the four manuscripts, all numerical results in this series are classified into one of three categories (see also Table 6, column "Type"):

- **Type A — Algebraic:** The value follows solely from φ and π through the Level 1–3 chain in Table 6. No observational data enters the computation. These results are *prior-free predictions.*
- **Type M — MCMC-validated:** The value is the median of the posterior distribution from the 100-walker `emcee` pipeline described in Paper 2, using DESI DR2 BAO, Planck 2018 priors, and galaxy-cluster mass data. These results reflect the empirical range accessible within the SSEE parameter family.
- **Type P — Prior-dependent:** The value depends on an external dataset anchor (e.g., Planck TT+TE+EE+lensing as a likelihood) that is not internally derived by the SSEE framework. Type P results quantify the consistency of SSEE with standard cosmological datasets; they are not Type A predictions.

The distinction is particularly important for the ΔBIC comparisons in Papers 2 and 3. In Paper 2, the $\Delta\text{BIC}=-5.55$ (dynamic sector) is a Type M result: it reflects the MCMC posterior relative to ΛCDM under identical priors. In Paper 3, the $\Delta\text{BIC}=-31.3$ (plik_lite likelihood, canonical $k=2$; aggressive $k=1$ gives -40.3) is a Type P result: it depends on the `plik_lite` likelihood as an external CMB anchor. These two BIC values are *not directly comparable* because they operate in different model spaces; the distinction is disclosed explicitly in Paper 3 §4.4.

D Algebraic Prior Space and Derivation Provenance

A legitimate concern in any zero-free-parameter model is whether the algebraic expressions for observables were selected *after* seeing the data (*post-hoc tuning*) or derived from a pre-committed algebraic structure. This appendix documents the derivation provenance of the key SSEE quantities to enable independent assessment.

Primary register — arithmetic derivation from $\{\varphi, \pi\}$

All seven structural constants of SSEE are derived from the two axioms in at most two layers of arithmetic (+, ×, integer scale). Four constants (Ω, β, K_v, M_v) are Level 1 — direct combinations of φ and π; three (KAL_0, P_{sc}, T_r) are Level 2 — using Level 1 constants (Ω, β) as intermediates:

Symbol	Name	Definition	Value	Layer
Ω	DNAV / Stability Metric	$\varphi + \pi$	4.7596	L1
β	BIAL / Base Coupling	$(\varphi + \pi)/2$	2.3798	L1
K_v	KRYSTOS / Structural Constraint	$2(\varphi + \pi)$	9.5192	L1
M_v	MIKAEL_V / Maximal Invariant	$3(\varphi + \pi)$	14.2788	L1
KAL_0	Structural Viscosity	$\beta + \pi$	5.5214	L2
P_{sc}	PYROS / Dynamical Scalar	$\Omega + \varphi$	6.3776	L2
T_r	TRIAL / Saturation Horizon	$3(\varphi + \beta)$	11.9935	L2

No constants were introduced *ad hoc*: all seven emerge from systematic enumeration of non-redundant, monotonically ordered arithmetic combinations of $\{\varphi, \pi\}$ up to two layers. The *closure criterion* is exhaustiveness within this family: $M_v = 3(\varphi + \pi)$ is the maximal element at Layer 1, and no additional Layer 2 combination yields a new structurally independent register. Products and integer powers of $\{\varphi, \pi\}$ are excluded from the primary register; they would introduce additional structure beyond the stated axioms.

AURA and MIRA — derivation from pre-existing constants

AURA was defined in the original SSEE-Unificado framework as

$$\mathrm{AURA} = \mathrm{BIAL} + \varphi = \beta + \varphi = \frac{\pi + \varphi}{2} + \varphi = \frac{3\varphi + \pi}{2} \approx 3.9978.$$

MIRA follows immediately as MIRA = AURA/2:

$$\mathrm{MIRA} = \frac{\beta + \varphi}{2} = \frac{3\varphi + \pi}{4} \approx 1.9989.$$

Both were defined algebraically before any comparison with Planck 2018 data. The numerical match $\mathrm{MIRA} \approx \Omega_{m,\mathrm{CMB}}/\Omega_{m,\mathrm{dyn}} = 0.3199/0.160 = 1.9994$ was *discovered* upon comparison, not engineered. Timestamp proof is provided by Zenodo Genesis 5.12 (DOI: 10.5281/zenodo.19679049, deposited 2026-01-28, prior to the Planck 2018 Ω_m confrontation in Paper 3).

Five exact consistency identities

The following identities hold exactly in $\{\varphi, \pi\}$ algebra and are verified to machine precision ($< 10^{-16}$ relative error):

$$w_0 = -\frac{\mathrm{AURA}}{\Omega_{\mathrm{DNAV}}} = -\frac{T_r}{M_v} \tag{52}$$

$$\Omega_{m,\mathrm{dyn}} = \frac{\pi - \varphi}{2(\varphi + \pi)} = 1 + w_0 \tag{53}$$

$$\mathrm{MIRA} = |w_0| \cdot \beta \tag{54}$$

$$\mathrm{KAL}_0\big|_{\varphi \leftrightarrow \pi} = \mathrm{AURA} \tag{55}$$

$$\mathrm{AURA}\big|_{\varphi \leftrightarrow \pi} = \mathrm{KAL}_0 \tag{56}$$

Identity (55)–(56) is the deepest structural result: under the discrete exchange $\varphi \leftrightarrow \pi$, the scalar-kinetic constant $\mathrm{KAL}_0 = (\varphi + 3\pi)/2$ maps *exactly* to the photon-sector constant $\mathrm{AURA} = (3\varphi + \pi)/2$, and vice versa. This discrete duality is the algebraic origin of the disformal coupling $\beta_c = -\mathrm{AURA}$ derived in Paper 7 §2.2: switching the $\varphi \leftrightarrow \pi$ roles between the two sectors forces $\beta_c = -\mathrm{AURA}$ as the unique consistent value. The open question of whether this discrete symmetry is an exact symmetry of the full $P(X, \phi)$ action is catalogued as OP-7 (`OPEN_PROBLEMS.md`).

Spectral index $n_s = 1 - \varphi^{-7}$

The SSEE algebraic commitment is $n_s = 1 - \varphi^{-7}$, pre-committed as part of the register structure independently of any inflationary framework (Zenodo Genesis 5.12, deposited 2026-01-28, prior to all Planck comparisons). It is not derived from the α-attractor class [Starobinsky, 1980, Kallosh & Linde, 2013, Ferrara et al., 2013]; rather, compatibility with that class is established *a posteriori* in App. A §A.5: the two independent SSEE axioms ($n_s = 1 - \varphi^{-7}$ and $r = \varphi^{-10}$) determine $N = 2\varphi^7 \approx 58.07$ and $\alpha = \varphi^4/3$ exactly as a consistency check, not as a derivation. The Planck 2018 value [Planck Collaboration X, 2020] $n_s = 0.9649 \pm 0.0042$ is 0.2σ from the SSEE algebraic value; this comparison was made after the axiom was committed.

Baryon density $\Omega_b h^2 = (\pi - \varphi)/[3(\varphi + \pi)^2]$

The structural expression $(\pi - \varphi)/[3(\varphi + \pi)^2] = (\pi - \varphi)/H_0^{\mathrm{SSEE}}$ gives 0.02242, at 0.32σ from Planck 2018 (0.02237 ± 0.00015). This form is classified as a *retrodiction*: it was identified after the data (OP-1, `OPEN_PROBLEMS.md`), replacing the preliminary placeholder $3(\pi - \varphi)/200 = 0.02285$ (3.2σ), in which the factor 200 was not an algebraic $\{\varphi, \pi\}$ quantity. No parameter is fitted: the denominator $3(\varphi + \pi)^2$ is the algebraic Hubble scale H_0^{SSEE}.